

Credit Score Analysis and Prediction

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1. Abstract

Our project focused on a credit score classification dataset. We aimed to develop a Machine Learning model that could predict credit score ratings for individuals. We obtained our data from Kaggle (a publicly available data source) and did light cleaning, feature encoding, and selection on the data. We evaluated multiple models, eventually settling on an XGBoost classifier due to its outperformance of other considered models. This model was able to have some significant predictive power of an individual's credit score rating based on their financial characteristics and credit history.

2. Introduction

In our project we aimed to develop a Machine Learning model that could predict a candidate's credit score rating. We tested various

models on the data and validated models with traditional ML performance metrics. (Accuracy, Precision, Recall, Punctuation F1, ROC and AUC, etc...) Data cleaning and feature selection were performed. We had multiple motivations for pursuing this project and potential applications for this research. This research is applicable in the finance and credit industry, or any industry that uses credit scores. The research is also valuable to any consumer looking to build or improve their credit score.

3) Proposed Methods

The models we considered for use in this project were a Multinomial Logistic Regression model and an XGBoost model. As stated previously we will compare these models using traditional ML model evaluation metrics such as Accuracy, Precision, Recall, Punctuation F1, ROC and AUC.

Multinomial Logistic Regression is a generalization of traditional Logistic Regression to fit classification problems with more than two outcome classes. In our case, this would be the credit score rating. This is given as either “Good”, “Fair” or “Poor.” This is accomplished by effectively running K independent binary Logistic Regression models, where K is the total number of classes. This is a very flexible model by nature, but is built off of a very interpretable model in Logistic Regression, giving us lots of explainable, predictive power.

XGBoost, short-hand for “eXtreme Gradient Boosting” is a model built off of an ensemble of decision trees. Through the gradient boosting logic, models are built sequentially with each new model built to correct the residuals (error) made by the previous models. Finally, the predictions of all models are combined to make the final prediction.

4) Experiments

4.1) Dataset, Data Cleaning & Feature Selection

Our data was publicly available via Kaggle. The data has 100,000 rows and 28 features. This includes identifying columns such as ID, Name, and Customer_ID.

During our project we had to do some light data cleaning. First, we had to do some data encoding and transforming, namely on features that were loaded in the data as non-numeric such as ‘Age’, ‘Monthly_Balance’ and some others. Additionally, in Age we noticed some irrational outliers in the data that we corrected with hard coding, Imposing a maximum of 100 and a minimum of 0. We then assessed missing values according to their columns. For missing NaN values in age we were able to substitute their value in Credit_History_Age_New plus the difference of the average Age value. We were able to do the inverse for Credit_History_Age_New’s missing values. We then identified all other columns containing null values, found the variables they were highly correlated to, and filled in the NaNs with predictions from a multiple linear regression model and a Logistic Regression model for appropriate columns. Finally, we encoded

Credit_Score and selected predictors based on their correlation to Credit_Score. The features/predictors we selected were:

"Num_Bank_Accounts", "Interest_Rate",
"Delay_from_due_date",
"Num_Credit_Inquiries", "Outstanding_Debt",
"Payment_of_Min_Amount",
"Credit_History_Age_New" and
"Monthly_Inhand_Salary."

4.2) Software

R Studio was used to process data, do analysis, fit and evaluate models. All work was done in Quatero ".qmd" files comparable to Python's Jupyter Notebooks.

4.3) Models

To start, a seed was set for random reproducibility. We ran our models with a 70% train size, a 15% validation size and a 15% test size. A Multinomial Logistic regression model was used via the Nnet library in R. The features used were the predictors we identified via feature selection. Accuracy, Precision, Recall, a Confusion Matrix, F1 Score and ROC curve with AUC value were generated from the model's predictions. Next, an XGBoost model

was also used via R's xgboost library. Our model used a learning rate of 0.1. This rate is lower, meaning the model training is more conservative but improves generalization. A higher selected value could have sped up training but risked overfitting which we wanted to avoid. We set a max depth of 6 and a minimum child weight of 1. Both of these values are still on the lower end to decrease the risk of overfitting. We set the subsample value equal to 0.8 meaning that 80% of the training data is used for each tree. We also set this as the colsample_bytree parameter meaning that 80% of the columns were randomly sampled for each tree. This decreases the depth and thus the risk of overfitting. The same model metrics as the Multinomial Logistic Regression Model were calculated as well for the XGBoost's predictions.

5) Results, Discussion

Our Multinomial Logistic Regression produced an AIC, Accuracy, Precision, Recall and F1 Score with values of 107760.7, 0.5758, 0.7785, 0.5866 and 0.6691 respectively. While

the AIC value was quite high, the other metrics were within the range of indicating that the model performed moderately well on the data. This was further supported by the ROC curve and the AUC value of 0.646, showing that the model did moderately better than random classification (see charts below)

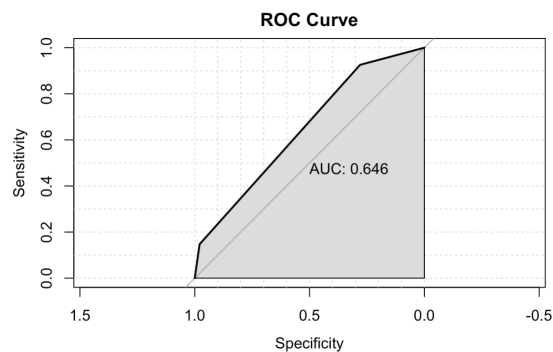


Fig 1: ROC Curve (Multinomial Logistic Model)

predictions_log	0	1	2
0	701	552	101
1	1750	5787	2328
2	53	1095	1493

Fig 2: Confusion Matrix (Logistic Model)

The XGBoost model performed much better with AIC, Accuracy, Precision, Recall and F1 Score values of 0.6939, 0.7344, 0.7363 and 0.7354 respectively. These values, along with the model's AUC of 0.755, show a sizable

improvement in predictive power over the Multinomial Logistic Regression model.

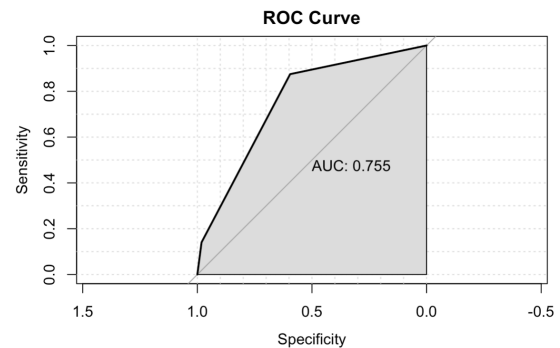


Fig 3: ROC Curve (XGBoost Model)

predictions_xgb	0	1	2
0	1490	930	267
1	968	5460	987
2	46	1044	2668

Fig 4: Confusion Matrix (XGBoost Model)

For further research, we would like to suggest calculating SHAP values (shapely additive explanation values) for the data to further investigate feature importance.

6) Conclusions

We found that credit score rating is somewhat predictable given an Individual's number of owned bank accounts, the interest rates on their credit cards, the average number of days they delayed payments, the number of

times they applied for credit cards, their outstanding debt, age of their credit history and monthly take-home salary.

Within the context of the problem, all of these predictors make sense to be significant predictors of credit score rating. To further assess the directionality of these predictors, i.e., whether they had a positive or negative impact on the individual's credit score rating we would suggest SHAP values be used.

Our XGBoost model outperformed the Multinomial Logistic Regression on all fronts with high scoring showing some predictive power.

References:

Data:

Credit Score Classification (Kaggle.com)

Obtained via:

<https://www.kaggle.com/datasets/parisrohan/credit-score-classification?resource=download&select=train.csv>